

Whisker-based tactile flight for tiny drones

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Tiny flying robots hold great potential for search-and-rescue, safety inspections, and environmental monitoring, but their small size and limited computational resources constrain onboard sensing capabilities. Inspired by animals such as rats and moles which rely on lightweight whiskers to navigate and perceive their surroundings through touch, we present a 3.2-gram whisker-based tactile sensing apparatus that enables tiny drones to perceive and interact with their environment through gentle physical contact, even in complete darkness. The apparatus employs barometers at the base of each whisker to estimate contact depth in flight, enabling obstacle localization while minimizing contact-induced destabilization. To compensate for sensor noise and drift during sustained contact, we develop a tactile depth estimation pipeline that achieves millimeter-scale depth estimation accuracy. Together, these innovations enable tiny drones to autonomously avoid obstacles, contour surfaces, and explore confined spaces, guided by onboard tactile sensing across both rigid and soft environments. Running entirely onboard a microcontroller with just 192 KB of memory, our system demonstrates autonomous tactile flight across various scenarios. This bio-inspired approach extends perception for mobile robots beyond vision, opening new possibilities for autonomous operations in visually degraded and GPS-denied environments.

Navigating complex environments is a fundamental skill for autonomous robots. However, tiny flying robots weighing under 100 grams face extreme limitations in sensing, computation, and power, making high-performance autonomy a major challenge^{1,2}. In low-visibility conditions such as smoke-filled buildings, dark caves, or areas with transparent or reflective obstacles, conventional sensors like cameras, LiDAR, or other rangefinders fail. These failures compromise both safety and mission success, particularly in high-stakes applications like search and rescue³, inspection⁴, or exploration⁵. To address these challenges, we introduce a tactile perception framework that enables vision-independent autonomous navigation for tiny flying robots and other mobile robots. We present a 3.2-gram whisker-based tactile sensor array inspired by rats' vibrissae, designed to operate entirely onboard, under strict constraints on payload and computation. Mounted on a palm-sized drone, our system uses MEMS barometer-based whisker sensors to accurately gather environmental information through touch. These robotic whiskers serve as lightweight, high-precision tactile receptors that allow drones to sense and respond to

the surroundings in real time, even when visual and proximity cues are unavailable.

The biological inspiration stems from animals such as rodents, which navigate narrow, dark, and complex spaces using whiskers that detect contact forces with minimal pressure⁶. Translating this capability to aerial robots, we use flexible artificial whiskers that generate differential pressure signals upon contact. These signals are processed to infer tactile depth, supporting robust navigation behaviors such as obstacle avoidance, surface following, and tactile mapping.

In contrast to traditional contact-based methods that rely solely on binary contact detection or passive bumpers^{7–12}, our whisker-based system provides rich and continuous tactile feedback with explicit tactile localization information, enabling non-intrusive, precise, and complex aerial interaction with the environment. The whiskers' low interaction force ensures safe contact with delicate or lightweight obstacles, while sliding contact along the whisker shaft further enhances spatial resolution and enriches tactile data.

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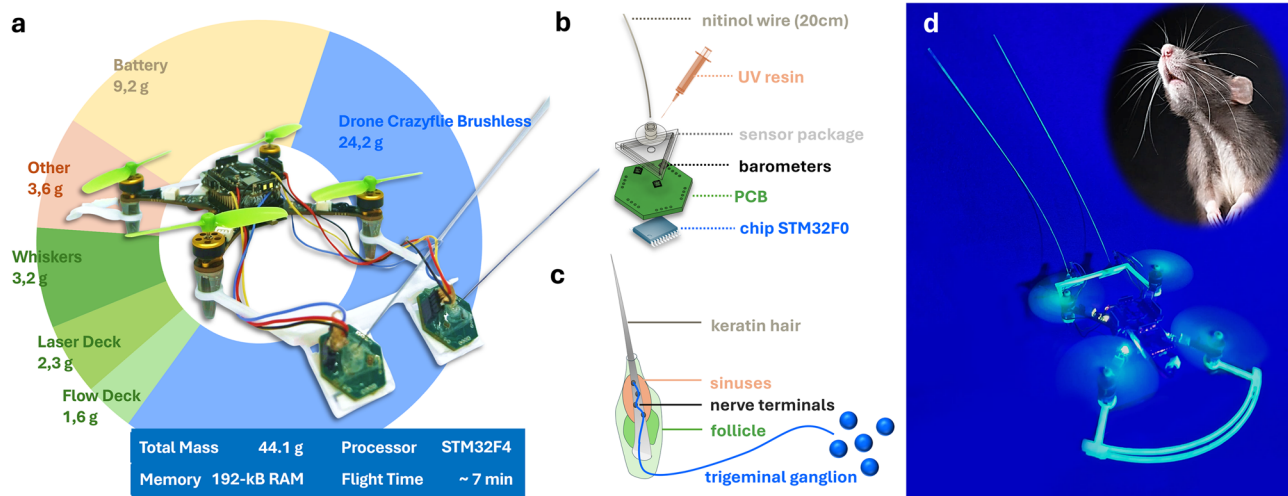


Fig. 1 | Design and biological inspiration of the whiskered drone. **a** The whiskered drone platform considered here: a tiny, 44.1-g Crazyflie Brushless drone with 2 artificial whiskers. **b** Structural diagram of the whisker sensor, the STM32F0 is only used for data acquisition and transmission for the three barometers and for

temperature compensation. **c** A simplified structural diagram of the rat whisker sensing mechanism³¹ (Color-Coded to Match **(b)**). **d** Whiskered drone wall sweeping and rat whisker sensing in darkness.

Previous work on contact-based control in drones has focused primarily on large aerial platforms equipped with robotic arms for manipulation^{13–17}, which are unsuitable for micro aerial vehicles due to stricter size and energy demands. Other efforts involving artificial whiskers have been limited to ground robots^{18–24} or to airflow-based sensing on drones without physical contact^{25,26}. Additionally, aerodynamic-based proximity sensing methods such as wall and ground effect detection^{27–29} offer limited precision and are sensitive to flight dynamics. Our work demonstrates whisker-based tactile exploration and navigation performed entirely onboard a tiny drone, with touch serving as the primary sensing modality for exploration and navigation.

To enable this, we develop and optimize lightweight whisker-like tactile sensors for tiny aerial robots, using force-aware structural design to minimize destabilizing torques. Complementing the hardware, we introduce a tactile signal processing pipeline that includes a Tactile Drift Online Recurrent Compensation (TDORC) algorithm, which corrects for drift caused by convective airflow, suppresses sensor ringing, and mitigates hysteretic post-contact recovery in the signal. A depth estimation model converts tactile signals into millimeter-precision proximity data, which is then integrated with the drone's motion model using a Kalman filter. This fusion ensures reliable perception even in dynamic conditions with sensor noise and latency. Despite this resource-constrained hardware, the system performs fully autonomous tactile flight—including real-time sensing, processing, decision-making, and control—without relying on external computation or localization infrastructure. The resulting tactile behaviors are complex and adaptive. We demonstrate that the drone can execute surface contour following in complete darkness, as well as navigate and map environments using only tactile information. In enclosed spaces, wall-following combined with active perception allows the drone to construct a tactile-based environmental representation. Notably, the entire tactile perception, navigation, and exploration pipeline operates onboard an STM32F405 microcontroller with only 192 KB of RAM, of which just 34 KB are used by our algorithms.

This work demonstrates the potential for robust, bio-inspired autonomy in small aerial robots, particularly in environments where vision fails. By shifting from passive sensing to active, touch-based exploration, we empower micro air vehicles to feel their way through the world—much like whiskered nocturnal animals do.

Results

A tiny drone with whiskers

We begin by detailing the construction of the whiskered drone platform, which involves designing and fabricating the whiskers, selecting the appropriate drone, and determining a suitable whisker placement angle. The system overview of our whiskered drone is illustrated in Fig. 1a. The concept, design, sensing principles, and fabrication of the whiskers build upon our previous work³⁰ with slight modifications, see Supplementary Materials. The structure of our whisker sensor is illustrated in Fig. 1b. Similar to the biological whisker shown in Fig. 1c³¹, it consists of four main components: (1) Whisker shaft: we employ a 0.4 mm diameter, 200 mm length Nitinol wire, pliant enough to enable subtle touch interactions; (2) Follicle-sinus complex (FSC): to transmit and amplify whisker motion while ensuring a flush connection with the mechanoreceptors, we employ a 3D-printed transparent package filled with UV resin to house the whisker shaft; (3) Mechanoreceptors: we use three barometers (BMP390), which measure pressure changes caused by the FSC unit during whisker bending; and (4) Neural transmission: sensory data is transmitted from the STM32F0 on the whisker PCB to the STM32F4 on the flight controller, enabling perception and control akin to biological processing in the animal brain. Similar to how a rat uses rhythmic whisking to scan and follow walls in the dark, our whiskered drone performs tactile sweeps along surfaces to explore its surroundings (Fig. 1d).

The whisker placement and the contact location in flight critically affect drone stability³². Despite generating only milli-Newton forces³⁰, the long force arm and limited thrust make the drone susceptible to yaw moments during wall sweeping, inducing instability. We used forward-facing whiskers aligned with the pitch plane (Fig. 2a) to support two-whisker wall-angle tracking while preventing the tips from catching on surfaces. To determine a suitable placement angle, we conducted theoretical and experimental analyses to evaluate how different angles affect yaw-inducing forces.

To evaluate the effect of whisker placement, the drone performed controlled sweeping motions along a wall (Supplementary Movie 1). We categorized placement angles in the pitch planes into three types: negative angles, 0°, and positive angles. The 0° placement was immediately ruled out due to whisker buckling and irregular signal outputs. Introducing a slight yaw angle in this configuration to maintain wall contact results in a normal force N , bending the whisker in the yaw plane, and a friction force f_N during sweeping; together, they

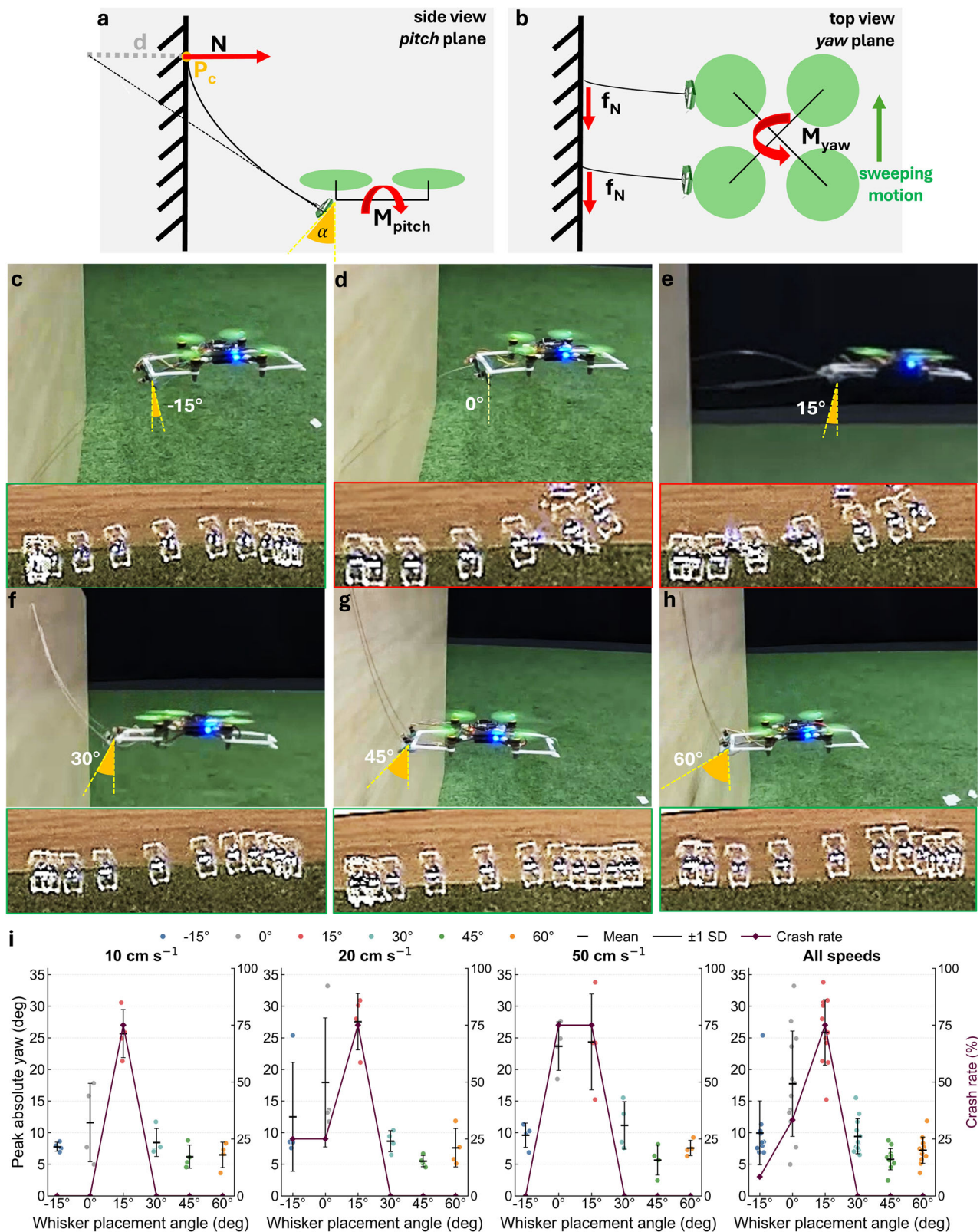


Fig. 2 | Effect of the whisker placement angle on the drone's stability during wall-following interactions. a, b Schematic illustration of forces and moments present on the drone during whisker-based wall sweeping at the positive placement angle. While in contact with a vertical obstacle, the whisker generates a normal force N , which induces a friction force f_N . The whisker's placement angle α influences the resulting N and moment arm, and thus the drone's pitch M_{pitch} and yaw M_{yaw} moments. **c–h** Experimental setup, whisker deflection, and sweeping motion of the drone with whiskers mounted at -15° , 0° , 15° , 30° , 45° , and 60° in the pitch

plane. Due to gravity, the whisker naturally tilts downward by approximately 10° from the preset angle. In the top-view clips we can observe sweeping behavior at 0.5 m/s : crashes primarily occur due to the drone's yaw deviation and inability to compensate for M_{yaw} . **i** Peak absolute yaw deviation and crash rate for the six placement angles at sweeping velocities of 0.1 , 0.2 , and 0.5 m/s . Each angle-velocity condition was repeated four times. The 45° placement yielded the smallest yaw deviation across all speeds, as well as no crashes.

induce a high yaw moment M_{yaw} , thereby compromising stability during wall sweeping. Negative and positive angles reduce yaw disturbance, as $f_N \ll N$, while the normal force primarily generates a pitch-plane moment M_N , which is easier to compensate (Fig. 2a, b). Between the two viable configurations, a positive angle is preferable. Forward-mounted whiskers generate a counterclockwise pitching moment M_w due to the forward shift of the center of mass. During contact, the normal force N produces a clockwise moment M_N that partially offsets M_w , thereby reducing the overall pitching moment M_{pitch} during contact interaction and reducing the need for additional rear counterweight. In contrast, negative angles reinforce M_w , increasing M_{pitch} and destabilizing the drone. Pseudo Rigid-Body Modeling (PRBM) shows that as the placement angle increases, the normal force N decreases for the same contact depth d at a static point P_c , reducing friction force $f_N \propto N$. From this, we derived a physical model relating N and d (“Methods”). This analysis suggests that positive whisker angles should improve interaction stability, although excessively large angles also reduce the effective forward sensing range.

For real-world validation, we conducted experiments on the Crazyflie 2.1 Brushless platform, a 44.1-g drone equipped with two whiskers. We evaluated six whisker placement angles (-15° , 0° , 15° , 30° , 45° , and 60°) at a contact depth of $d = 8\text{ cm}$ while sweeping along a smooth wall at 0.1, 0.2, and 0.5 m/s (Fig. 2c–h). Each angle–velocity condition was repeated four times, resulting in 72 trials in total. We quantified stability using the peak absolute yaw deviation during contact and the crash rate (Fig. 2i). Across all speeds, the 15° configuration showed the largest yaw deviation and the highest crash rate, consistent with the whisker drooping toward a near- 0° configuration under gravity. The 0° configuration also showed large yaw excursions and frequent crashes. The -15° configuration produced one crash, consistent with the additional adverse pitch moment predicted by the analysis above. In contrast, the 30° , 45° , and 60° configurations completed all trials without crashes. Among these, the 45° placement yielded the smallest mean peak yaw deviation across speeds, indicating the most stable wall interaction (Fig. 2i). Although the 60° configuration also remained crash-free, it did not further reduce yaw deviation and shortened the effective forward sensing range of the whiskers. One possible reason is that, at 60° , the shortened forward reach brings the drone body closer to the wall at the tested contact depth, which may introduce additional aerodynamic-related disturbances. We therefore selected 45° as the whisker placement angle because it provided the best balance between interaction stability and usable sensing range. Preliminary experimental observations obtained on the Crazyflie 2.1 Brushed platform are included in the Supplementary Materials. Video compilations are provided in Supplementary Movies 1 and 10.

Online tactile drift correction

Pressure drift is a common issue in barometers, primarily caused by the combined effects of temperature fluctuations and partial inelastic strain recovery of the gel when subjected to significant force. On platforms like drones, these temperature variations are particularly pronounced due to convective cooling from the propeller-generated airflow. Prior methods³³ adjusted temperature compensation coefficients to reduce drift, but this requires extensive calibration and is limited by sensor-to-sensor variability. Moreover, false positives (FP) in contact detection may still mislead control. Alternative approaches, such as bandpass filtering alone^{22,34}, are unsuitable here because a high lower cutoff would suppress essential low-frequency tactile signals at the whisker tip.

We propose the TDORC algorithm to address three noise sources in tactile sensing: temperature-induced signal drift, post-contact hysteretic recovery, and vibration-induced ringing (Fig. 3b, yellow box). TDORC estimates temperature drift via first-order least squares within a sliding window ($n = 100$), updating it continuously during free flight,

unlike Tactile Drift Online Compensation (TDOC), which calibrates only once at takeoff. The estimated drift is subtracted from the raw data, followed by band-pass filtering to suppress high-frequency ringing and mitigate delayed recovery toward baseline after contact loss. To avoid corrupting the drift model with contact forces, calibration updates are applied only when no contacts are detected (e.g., $S_p^{up} \leq T$). This method runs in real time and integrates seamlessly into our system (Fig. 3a). Further details are in “Methods”.

To evaluate TDORC, we conducted free-flight experiments using the six-channel whisker sensors (Fig. 4a and Supplementary Movie 2). Ideally, all channels should read zero; we computed the mean and variance of signals and compared TDORC with raw and bandpass-filtered signals. TDORC effectively suppressed drift, keeping mean and variance closest to zero (Fig. 4b) and dynamically adapted to changing environmental conditions, unlike TDOC, which calibrates only once at takeoff. Violin plots show a single-peaked distribution with no signals exceeding the contact threshold, minimizing FP (Fig. 4c). The false positive rate (FPR) was calculated as $FPR = \frac{FP}{FP+TN}$, where true negatives (TN) corresponds to free-flight segments. Using a threshold of 20, the FPRs were 38.24% for bandpass filtering, 12.23% for TDOC, and 0% for TDORC.

Whisker-based tactile depth estimation

Contacts on the whisker sensor induce bending moments at the base, which we use to estimate the forward-facing distance d to obstacles. By projecting along the drone’s forward direction, the estimated depth directly corresponds to the distance to obstacles, ensuring safe flight. While in principle PRBM could calculate d from base moments, this requires calibrating the outputs of three barometers to contact forces using expensive high-precision Force/Torque (F/T) sensors, and can only be done while stationary, impractical in dynamic flight.

To estimate tactile depth, we combine a neural network-based sensor model with a process model using a Kalman Filter. The network is trained on data collected while the drone performs sweeping maneuvers near walls, allowing it to capture dynamic effects such as wall aerodynamics, platform vibrations, propeller airflow, and the frictional behavior of the nitinol whiskers. The Kalman Filter then fuses these sensor predictions with the process model, producing temporally consistent depth estimates while reducing computational load compared to our previous time-series-based approach³⁰. Further implementation details are provided in the “Methods” section.

Tactile depth estimation datasets were collected under two conditions: a whiteboard wall (Dataset 1) and a glass wall (Dataset 2), each with 12 flights (9 for training, 3 for testing). Ground truth (GT) positions were recorded with OptiTrack, and a ToF laser sensor was used for comparison. Details of the data collection protocol are provided in “Methods” (Fig. 5a, d and Supplementary Movie 3).

We compare our proposed method with depth estimation achieved using the sensor model alone, based on variants of the two versions of the process models and the MLP-based sensor model. Among the sensor models, we present only the MLP in the main text, as it outperforms the other tested baseline sensor models on Dataset 1. Detailed quantitative comparisons for all sensor models are provided in the Supplementary Materials, including evaluation metrics across trials (Table S1), hyperparameter search spaces and selected values (Table S2), and additional prediction, deviation, and reconstruction results (Fig. S6). The results from Dataset 1 show a clear improvement in prediction performance when integrating both versions of the process models, especially with the Kalman Filter (KF) full model. This enhancement is evident in whisker depth estimation, wall angle prediction, and surface reconstruction, with both MAE and RMSE metrics improving significantly (Table 1, Dataset 1). The reconstruction error is the minimum distance between the estimated contact points and the wall surface. The prediction error density plot (Fig. 5b) clearly shows the

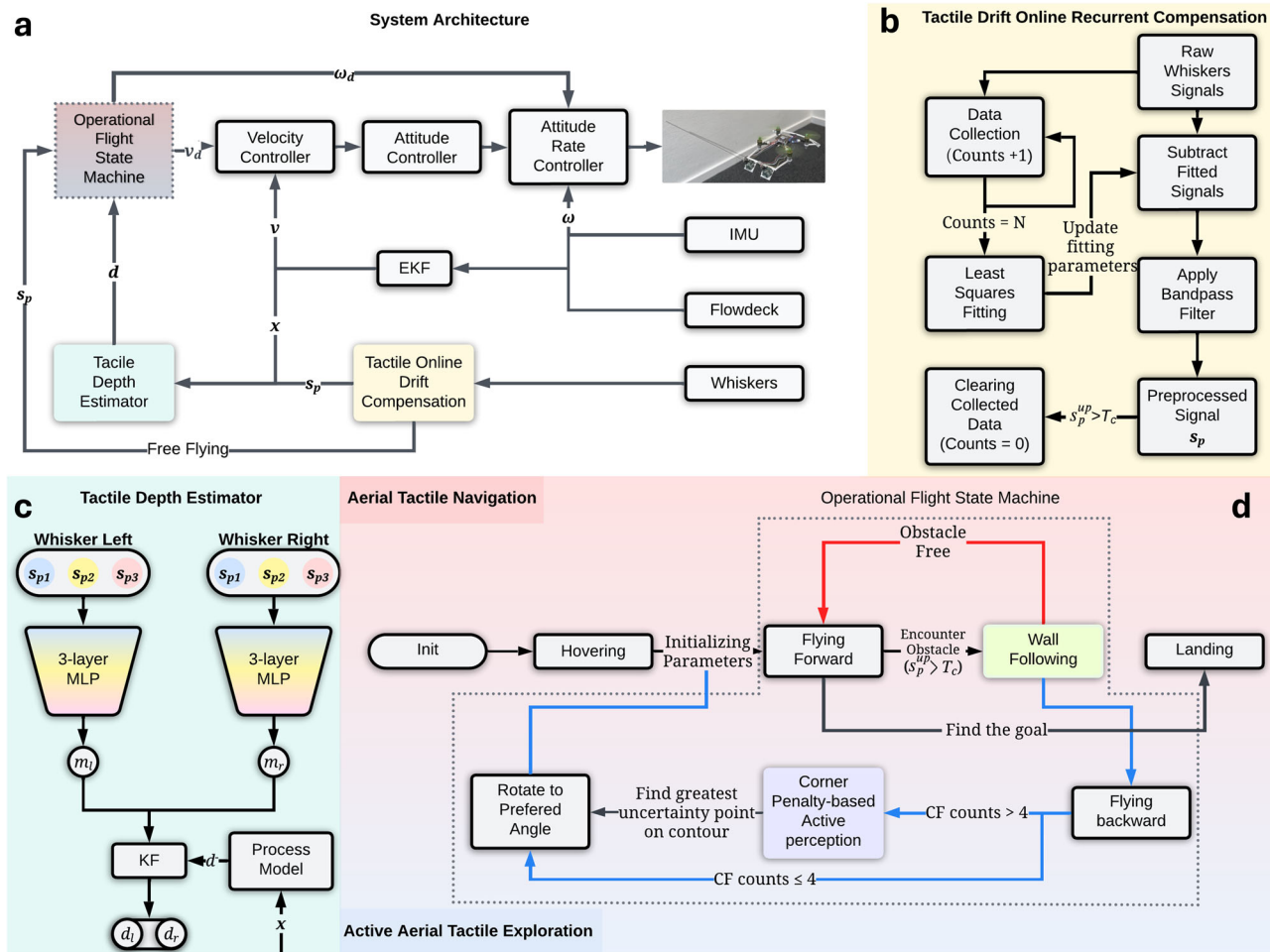


Fig. 3 | Overview of our whiskered drone system pipeline. a System architecture diagram. **b** The TODRC module for compensating tactile drift. **c** The tactile depth estimation module, which fuses the sensor model and process model to predict left and right whisker depths. **d** The aerial tactile navigation module (highlighted in red), based on wall-following, and the active aerial tactile exploration module

(highlighted in blue), which integrates wall-following and corner penalty-based active perception. The shared components used in both modules are represented with black arrows. For further details on wall-following and corner penalty-based active perception, please refer to the Supplementary Materials.

performance of the MLP + KF (Full Model), with the density closest to zero, followed by the MLP + KF (Simplified), and then the sensor model alone. The integration of both sensors' outputs, through the predicted orientation angle of the wall, significantly enhances the prediction performance of the left whisker, which experiences less stress due to the orientation change to the left caused by friction during the sweeping process. Figure S6a shows that the fusion process model helps mitigate sensor hysteresis effects, while deviation and reconstruction plots confirm that the fusion method brings predictions closer to the true values. Unlike the raw sensor response, the reconstructed trajectory better represents the actual object surface rather than simply following the drone's motion. Additionally, as shown in Fig. S6c, our whisker sensor's depth predictions closely match those of the ToF sensor, even on a white-board surface where the laser sensor excels.

For Dataset 2, the tactile depth estimation results exhibit similar trends. Our proposed fusion method consistently outperforms the MLP model alone (Table 1, Dataset 2), improving the accuracy of both whiskers and mitigating sensor hysteresis (Figs. 5e and S6b). Depth predictions remain robust even when using less precise state estimation (FlowDeck), and whisker-based estimates outperform the ToF sensor on transparent glass, where laser measurement noise is higher (Fig. 5f).

Aerial tactile flight through invisible walls

Tactile navigation methods in aerial and ground robots have predominantly relied on wall-following strategies rather than discrete and sparse navigation methods, such as whisker tapping. Studies on wheeled ground robots, such as refs. 35¹⁹, implemented wall-following tactile navigation using whisker sensors. More recent approaches on drones make use of robotic fingers¹⁴ and mechanical bumpers³⁶ for a wall-following navigation strategy. Wall following is particularly advantageous as it provides smoother navigation, continuous environmental perception, and reduced impact forces compared to discrete-contact-based methods, which rely on intermittent collisions to gather spatial information. Approaches like whisker tapping can introduce unstable motion, structural stress, and inefficient exploration due to repeated impacts, making them less suitable for delicate aerial systems. However, wall following requires precise control³⁷, but the whisker sensor eases this by acting like a range sensor that enables stable, non-intrusive contact. Its extended sensing range combines the robustness of tactile sensing with the perception range of distance sensors, allowing for greater control tolerance. We designed two experiments to validate the effectiveness of our proposed whisker-based aerial tactile navigation strategy. The first experiment evaluated the drone's ability to navigate unknown environments through three transparent walls made out of acrylic panels (Fig. 6a). The drone fused

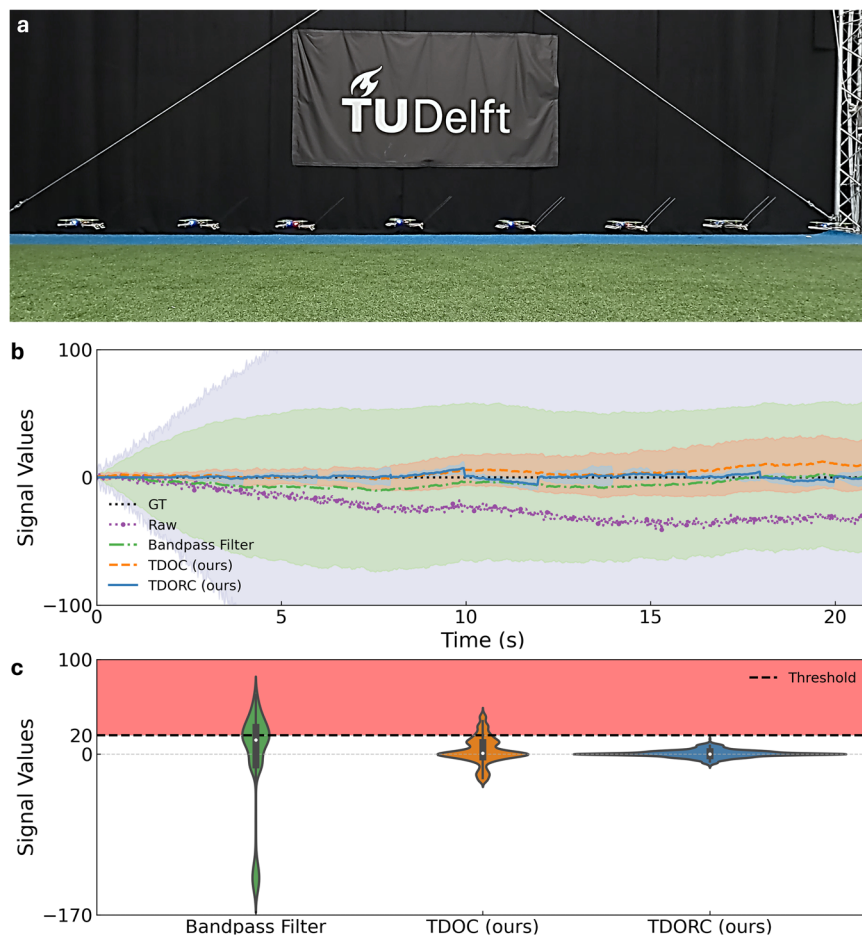


Fig. 4 | Whisker sensor drift compensation during free flight. **a** Whiskered drone performing a free flight with TDORC onboard. **b** The mean and standard deviation bands of the six-channel barometer signals for raw measurements, bandpass-filtered signals, TDOC, and TDORC demonstrate that our proposed methods effectively suppresses signal drift. **c** Violin plots showing the whisker signal

distributions across all channels and time for the bandpass filter, TDOC, and TDORC. The desired distribution is close to zero to maintain high sensitivity for low-pressure measurements. Our proposed methods minimize FP during free flight.

IMU and optic flow data for improved state estimation. It flew forward at $V_{\max} = 20$ cm/s, switching to wall-following mode upon detecting obstacles. The drone continued sweeping through the obstacles encountered and eventually landed. Figure 6b illustrates the trajectory, which consists of the absolute position and orientation of the drone across five independent trials conducted in this setup. Each trajectory is color-coded, with the drone's orientation visualized accordingly. Across all trials, the drone successfully avoided obstacles and reached the designated target location, demonstrating the robustness of the proposed aerial tactile navigation strategy. The perception and control framework ensures that the drone maintains a controlled standoff distance from the surface, constrained within an empirically determined range of 60–100 mm. As shown in Fig. 6c, the estimated contact depths remained within the threshold band, confirming the accuracy of the tactile depth estimator and the effectiveness of the control strategy. In Fig. 6d, we present the absolute orientation of the drone. The three walls were oriented at -90.76° , -91.34° , and -90.77° , respectively. Although each wall has a distinct absolute orientation (normal to the drone's orientation), for clarity, we display only the average of -0.95° (dashed line) in the figure. The drone's yaw fluctuated slightly around this reference, with a tendency to overshoot due to friction during wall contact, but corrections ensured that alignment with the wall was maintained.

In the second experiment, we introduce different wall orientations, i.e., angled in-and-out with respect to the drone heading

(Fig. 6e). The depth threshold was adjusted to 60–105 mm to accommodate these angles. All five trials succeeded in reaching the target (Fig. 6f), confirming the system's adaptability. The drone dynamically rotated clockwise or counterclockwise to align with the local wall orientation, effectively tracking the varying wall orientations. As seen in Fig. 6g, estimated depths fluctuated more frequently around the thresholds due to wall angle changes, resulting in minor oscillations, but the system maintained accurate distance regulation. Similar to the earlier case, Fig. 6h shows only the drone's ideal absolute headings-perpendicular to the walls-for clarity: -1° for the first and third (black dashed line), and 24.3° for the second (gray dashed line), while the actual wall orientations were -90.90° , -65.71° , and -91.15° , respectively. The drone maintained orientation close to these references, with slight deviations, mainly due to interaction friction. In one trial, a notable counterclockwise yaw drift occurred after reaching the destination, caused by signal drift in the left whisker sensor. Despite this, the drone consistently preserved near-perpendicular alignment during wall following. Video compilations of all the flights are available in Supplementary Movie 4, which also demonstrates the laser performance: in addition to the noise and intermittent failures shown in Fig. 5f, the laser completely fails when the drone is not perpendicular to the transparent walls.

In addition, we evaluate the robustness of the proposed tactile processing pipeline under post-contact oscillations, which occur when

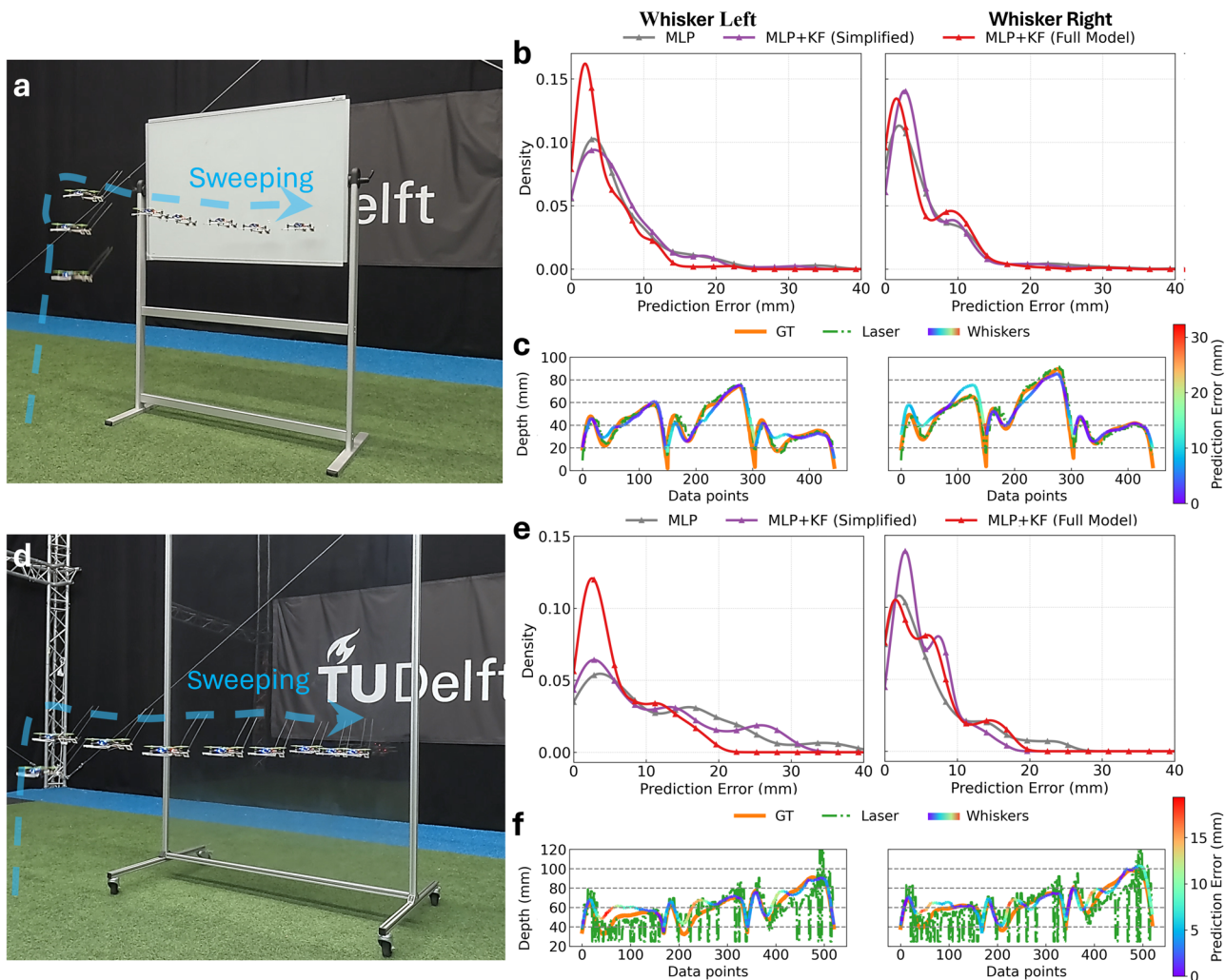


Fig. 5 | Whisker-based tactile depth estimation evaluated on the test set. a Data collection on a rigid panel (Dataset 1). **b** Prediction error density across models for Dataset 1. **c** Predicted depth from MLP + KF (Full Model) vs. GT and laser on Dataset 1. **d** Data collection on a rigid and transparent panel (Dataset 2). **e** Prediction error density across models for Dataset 2. **f** Predicted depth from MLP + KF (Full Model) vs. GT and laser on Dataset 2.

Table 1 Evaluation metrics for different models across two datasets: Dataset 1 (whiteboard) and Dataset 2 (glass wall)								
Dataset 1—collected by flying close to a whiteboard								
Model	Whisker left		Whisker right		Orientation		Reconstruction	
	(mm)		(mm)		(°)		(mm)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
MLP	6.40	9.07	5.36	8.08	6.18	7.72	5.34	8.06
MLP + KF (Simplified)	6.19	8.35	5.13	7.09	5.96	7.36	4.73	6.99
MLP + KF (Full Model)	4.23	5.69	4.72	6.75	5.55	6.84	4.12	5.75
Dataset 2—collected by flying close to a glass wall								
MLP	11.33	14.69	5.78	8.15	11.41	14.79	8.13	9.87
MLP + KF (Simplified)	10.09	13.27	5.11	6.24	10.01	13.44	8.49	10.25
MLP + KF (Full Model)	5.91	7.68	5.38	7.03	5.06	6.60	5.71	7.06

The table reports MAE and RMSE for each model across different estimation categories evaluated on the test set. Bold values denote the best-performing method.

the whisker slides off an obstacle. These oscillations are caused by whisker rebound and transient drone vibrations, and introduce high-frequency disturbances in the raw tactile signals. The proposed TDORC module effectively suppresses these oscillations before they are used for contact detection and depth estimation. Detailed results and analysis are provided in the Supplementary Material (Fig. S7).

We further evaluated the system under violations of the locally planar and static-obstacle assumptions, including curved surfaces and a moving surface. The system remained capable of wall following under moderate curvature and object motion, although oscillations increased and object reconstruction quality degraded in more dynamic cases. Detailed results are provided in Supplementary Fig. S9 and Supplementary Movie 8.

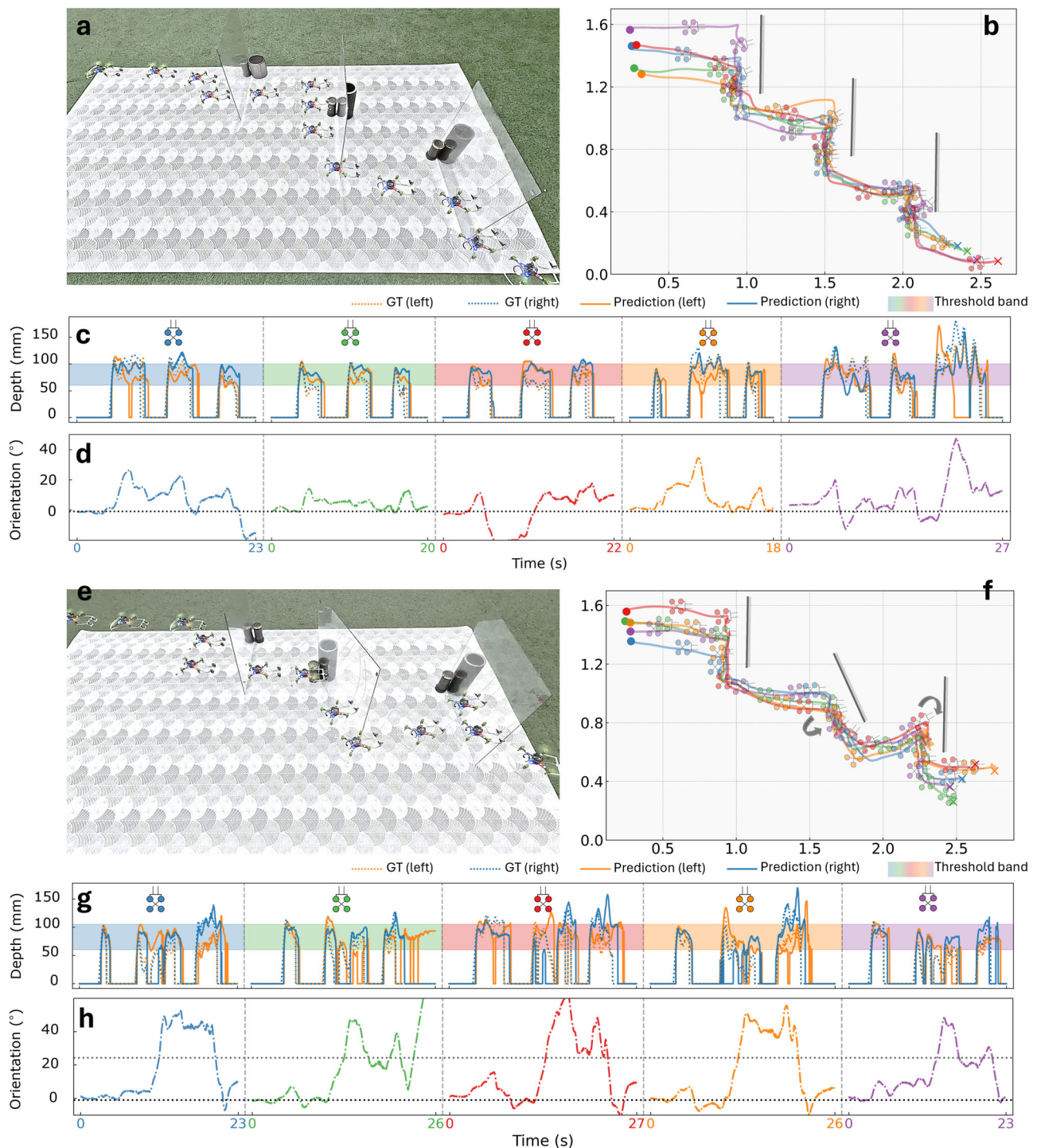


Fig. 6 | Experimental results of aerial tactile navigation through multiple invisible walls. a Whiskered drone successfully navigating three parallel glass walls. **b** Trajectories of the drone across five independent trials in the first wall setup, with absolute position and orientation visualized. **c** Comparison of true and estimated contact depths with the corresponding threshold bands in the first setup. **d** Absolute orientation of the drone in the first setup. **e** Whiskered drone

successfully navigating three differently-oriented glass walls. **f** Trajectories of the drone in five trials in the second setup. **g** Comparison of true and estimated contact depths with the corresponding threshold bands in the second setup. **h** Absolute orientation of the drone in the second setup. The GT was derived from the initial wall positions and the drone's relative pose, yet its accuracy can be affected by wall disturbances during interaction.

Active aerial tactile exploration in unknown confined environments

Animals and humans have developed various strategies to explore unknown environments in darkness. Nocturnal animals like rodents use whiskers to follow walls and detect obstacles, while blind humans tap canes to construct mental maps. These strategies have been applied to wheeled robots^{35,38} and manipulators²². Inspired by these

biological principles, we present what we believe is the first method for active aerial tactile exploration, combining wall-following for robust perception with Gaussian Process Implicit Surfaces (GPIS)-based active exploration for efficient understanding.

Wall-following offers stable tactile feedback but has limitations: it may trap the drone in corners, miss lateral obstacles due to front-facing whiskers, or follow unnecessary contours. Tapping-based

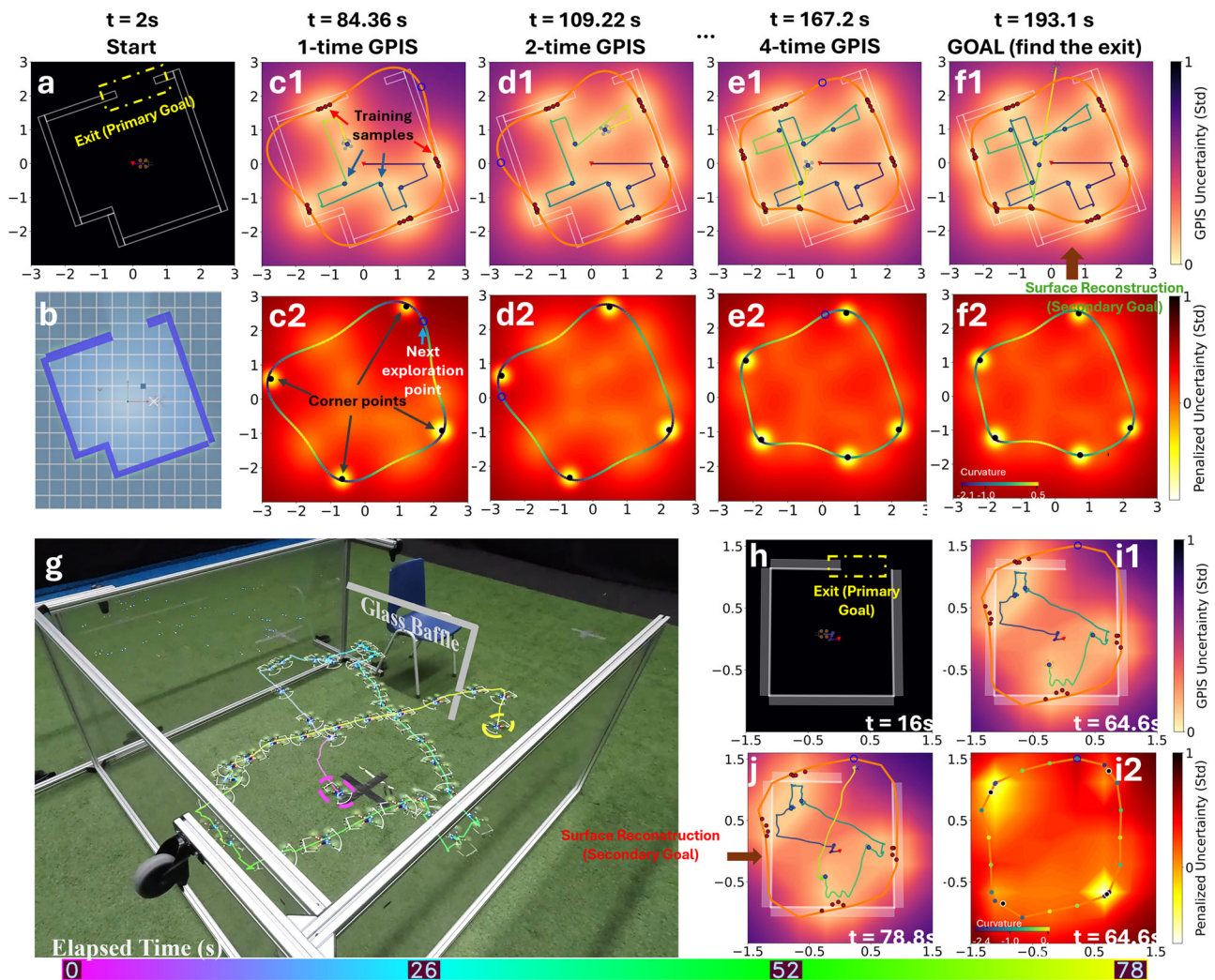


Fig. 7 | Simulation and real-world results of active aerial tactile exploration. **a** Initial unexplored environment with maximum uncertainty. **b** Simulation in ISAAC SIM. **c1, d1, e1, f1** Successive GPIS applications after collecting training data (red: surface; blue: interior), showing reduced uncertainty and improved shape reconstruction. The drone's trajectory (rainbow line), reconstructed contours (orange), and real-time state (drone icon) are shown. The exit is reached after the fourth GPIS. **c2, d2, e2, f2** Curvature analysis of extracted contours: brighter colors indicate higher curvature. Identified convex corners (white dots) are penalized, and

the next exploration target is selected (blue hollow circle) based on updated uncertainty. **g** Real-world trajectory with time-color-coded path; start and landing points marked. **h** Initial environment map with maximum uncertainty. **i1** Initial exploration in four directions to gather GPIS data, with corresponding trajectory and reconstruction. **i2** Real-world curvature analysis guiding target selection. **j** Final reconstructed map with successful exit navigation; minor deviations are due to odometry drift.

methods^{39–43} provide only sparse data and require repositioning, making exploration inefficient. Unlike prior work^{22,44} on object surface exploration, our focus is on navigating enclosed spaces, introducing unique challenges, especially concave corners that would yield collisions. We address this by detecting high-curvature regions via GPIS and assigning corner penalties, guiding the drone toward uncertain yet safe areas, see “Methods”.

To minimize cost and interaction forces, the drone uses only two forward-facing whiskers, sufficient for wall-following. Our method is designed to achieve two key goals: exploring the environment and identifying the exit. As shown in Fig. 3d, the drone alternates between wall-following and GPIS-based uncertainty reduction. This corner-aware strategy ensures safe, efficient, and complete tactile exploration in unknown, confined spaces. Detailed implementation is described in the “Methods” section.

We conducted ten simulation trials in randomly rotated room orientations. The proposed method successfully located the exit in 9 out of 10 trials (90% success rate) within 5 min, achieving a mean

reconstruction accuracy of 0.26 ± 0.06 m (CD) and 0.14 ± 0.03 m (RMSD), with a mean travel distance of 25.38 ± 5.26 m.

To further evaluate the contribution of each component, we perform a systematic ablation study, as summarized in Table 2. The definitions of the baseline variants and evaluation metrics are provided in “Methods”. The full method consistently outperforms all ablated variants in both reconstruction accuracy and exploration efficiency. Removing the corner-penalty module significantly degrades performance, reducing the success rate to 50% and increasing both travel distance and sweeping count. Similarly, removing the sweeping behavior further deteriorates both mapping accuracy and efficiency. Random exploration performs the worst across all metrics, as it does not explicitly leverage the acquired observations to guide exploration. Failure cases are further analyzed in Fig. S10, which breaks down the outcomes into success, crashing, and timeout events. The results reveal distinct failure patterns across methods. Removing the corner-penalty module leads to a substantial increase in crashing events, which account for the majority of failures in this variant. In contrast,

Table 2 | Quantitative evaluation of active aerial tactile exploration

Method	Exit success	Reconstruction accuracy		Exploration efficiency	
	SR (%)↑	CD (m)↓	RMSD (m)↓	Distance (m)↓	Sweeping count↓
Ours (Full)	90	0.26 ± 0.06	0.14 ± 0.03	25.38 ± 5.26	7.00 ± 1.41
w/o Corner penalty	50	0.27 ± 0.08	0.16 ± 0.06	34.16 ± 4.48	8.60 ± 0.89
w/o Sweeping	40	0.33 ± 0.09	0.19 ± 0.07	32.60 ± 10.88	10.11 ± 3.48
Random exploration	10	0.36 ± 0.12	0.25 ± 0.13	33.86 ± 9.16	10.50 ± 3.00

The table reports exit success rate, reconstruction accuracy, and exploration efficiency for different methods, including ablation variants. Exit success is defined as successfully locating and navigating out of the environment within 5 min. Reconstruction accuracy is measured by Chamfer Distance (CD) and Root Mean Square Distance (RMSD). Exploration efficiency is measured by travel distance and sweeping count, and is computed only over trials that were not prematurely terminated by crashing events. Each method is evaluated over 10 trials with randomly rotated environment orientations. To ensure fair comparison, trials in which the drone exits the environment immediately without performing meaningful exploration are excluded. Bold values denote the best-performing method.

removing the sweeping behavior results in a higher proportion of timeout cases, suggesting inefficient exploration with limited environment coverage. Random exploration exhibits both frequent crashing and timeout events, reflecting the lack of an information-driven exploration strategy.

Additional qualitative results across all trials are provided in Supplementary Fig. S11, where we visualize the drone trajectories, reconstructed contours, GT layouts, and failure cases (indicated by red crosses for crashing events). These results further highlight the role of the corner-penalty mechanism in avoiding concave regions. Without this component, the drone frequently collides with sharp corners, as observed in both the w/o corner-penalty and random exploration variants. In contrast, the proposed method consistently steers away from such high-risk regions. We also observe that removing the sweeping behavior leads to noticeably less accurate reconstructions, as the lack of local coverage limits the acquisition of fine geometric details. Finally, we note a limitation of the current strategy: in some cases (e.g., Trial 8), the drone locates the exit before fully exploring the environment, resulting in incomplete reconstruction. This suggests that future work could incorporate an uncertainty-aware exploration strategy to prevent premature exit and enable the drone to return and continue exploring until sufficient environment coverage is achieved.

Figure 7 illustrates one successful exploration, which panel Fig. 7c1 shows the first application of GPIS to the uncertainty map after collecting training data in four orthogonal orientations (red dots: surface; blue dots: interior). Compared to Fig. 7a, uncertainty is significantly reduced in the explored regions. The drone's trajectory and current state are indicated by a small drone icon, with reconstructed contours in orange using the marching squares algorithm. Curvature analysis (Fig. 7c2) highlights high-curvature corners, which are penalized to guide the next exploration target (blue hollow circle). Subsequent GPIS applications and corner penalties (Fig. 7d1–f2) progressively refine the map and reduce uncertainty. For clarity, we omit the third GPIS application from the figure. The drone successfully located the exit after the fourth GPIS application.

To validate our method in real-world scenarios, we constructed a 2 m × 2 m room with glass walls and an 80 cm-wide exit. As part of the experimental setup, optical flow is used only for egomotion estimation and flight stabilization, while all environment perception and navigation decisions are driven by tactile sensing onboard. As shown in Fig. 7g, h, the drone successfully explored the environment and located the exit. It began by collecting GPIS training data in four orthogonal directions (Fig. 7i1) to build an initial environmental model. High-curvature corners were identified and penalized (Fig. 7i2), guiding the drone to explore in safer directions while progressively reducing uncertainty. At each step, the GPIS-based prediction and corner penalties updated the target exploration points, allowing the drone to navigate efficiently and avoid collisions. The final reconstructed map closely aligns with the actual room layout, with minor deviations due to odometry drift. Grid resolution was reduced (10 × 10) compared to

the simulation to save memory. Videos of the real-world exploration are available in Supplementary Movie 5.

Aerial tactile wall following in complete darkness

We evaluated the drone's ability to perform wall-following flight in complete darkness on both solid (Fig. 8a) and soft, textile surfaces (Fig. 8c). In these settings, we employ the aerial tactile navigation framework of Fig. 3, relying solely on the onboard IMU and a downward-facing ToF sensor for state estimation. Tactile depth was predicted only using MLP.

The drone took off and flew forward at $V_{\max} = 20$ cm/s. Upon detecting an obstacle ($T_c = 20$), it entered wall following mode, maintaining a safe standoff distance within a depth threshold band of 40–100 mm for approximately seven seconds before flying backward and landing. Although laser data was recorded, it cannot be considered GT due to sensor placement differences and unknown relative positioning. Nevertheless, depth predictions from both left and right whiskers closely tracked the trends in the laser data, indicating sufficient accuracy for closed-loop tactile navigation (Fig. 8b, d). Despite significant IMU drift, the drone was still able to maintain stable wall-following using reactive control. As shown in Supplementary Movie 6, the drone exhibited an increase in lateral velocity while transitioning from the first to the second obstacle during solid surface sweeping. This may have resulted from a momentary rightward acceleration when losing contact, which the IMU misinterpreted as relative stasis, prompting an excessive corrective command in that direction. Wall-following on soft objects proved more challenging due to higher surface friction and the presence of folds on the flag. These factors reduce the reliability of depth estimation and destabilize the controller, resulting in the oscillatory behavior observed in Fig. 8d. The oscillations highlight the controller's limited robustness under soft-surface conditions. Nevertheless, the system successfully completed wall-following on the flag without crashing—a task that remains difficult for other intrusive tactile sensors or aerial manipulators.

Discussion

In this work, we demonstrated a whisker-based tactile perception and control framework that enables sub-50 g drones to navigate and explore challenging environments where vision and traditional sensors are unreliable. By drawing inspiration from whiskered animals, we validate that continuous, non-intrusive tactile sensing is a viable alternative for real-time obstacle avoidance and environment mapping around invisible obstacles, but also for wall following in complete darkness. Our key innovations include the design of a lightweight, barometer-based whisker sensor optimized to minimize destabilizing forces, and a real-time signal processing pipeline incorporating the TDORC algorithm, neural network, and Kalman filtering to significantly enhance depth estimation accuracy by mitigating sensor drift and noise. We further developed an aerial tactile navigation and exploration framework that enables real-time surface contour following in complete darkness and efficient environmental navigation and

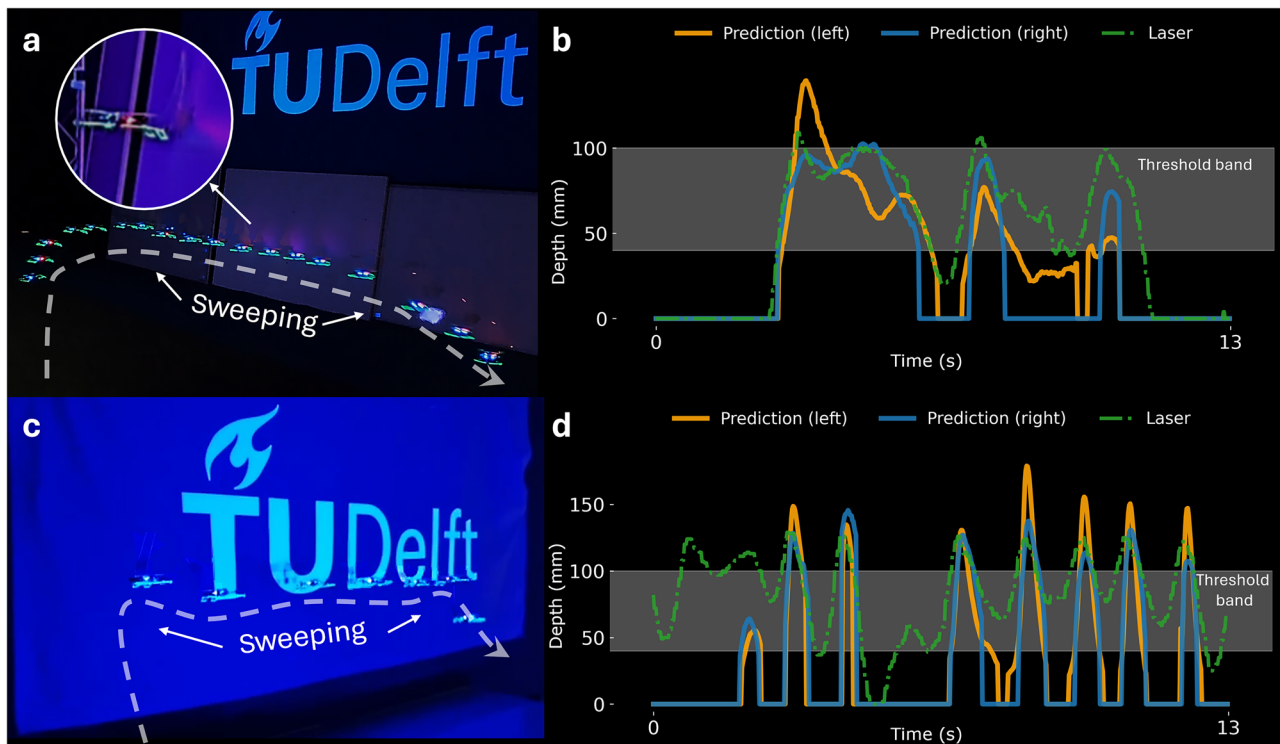


Fig. 8 | Wall-following-based aerial tactile navigation in complete darkness along solid and soft surfaces, using only the onboard IMU and a downward-facing ToF sensor for state estimation. a Sweeping along a rigid surface for 7 s. **b** Predicted depth from the whisker MLP model vs. laser measurements during

tactile flight in the dark, along the rigid surface. **c** Sweeping along a soft, textile surface for 7 s. **d** Predicted depth from the whisker MLP model vs. laser measurements during tactile flight in the dark, along the soft textile surface.

mapping around invisible obstacles. Importantly, the software stack operates entirely onboard with just 192 kB RAM on an STM32F405 microcontroller at 50 Hz, showcasing the feasibility of implementing bio-inspired tactile sensing on resource-constrained platforms without relying on external computation. This research highlights the potential of whisker-based tactile sensing as a transformative approach for tiny drones in challenging environments.

One limitation of our method is drift accumulation during extended periods of time in tactile sweeping, which can degrade whisker depth estimation, especially under disturbances that increase dynamic effects in aerial navigation. Incorporating time-series modeling to predict drift alongside tactile measurements may help mitigate this issue. Scalable fabrication and calibration remain open challenges, as barometric inconsistencies and small mechanical variations hinder direct model transfer across devices. In the current setup, achieving the best performance requires a short per-device recalibration procedure based on approximately 5–6 controlled sweeps with external ground-truth sensing, which takes about 15–20 min. Future transfer-learning approaches with brief multi-directional sweeps, ideally requiring little or no labeled data, may improve scalability.

Despite these limitations, our current work lays the foundation for further advances in autonomous aerial tactile exploration on drones. By leveraging bio-inspired tactile sensing, we show that tiny drones can perform robust and efficient exploration in complex, challenging environments. Our approach challenges the reliance on traditional vision-based and proximity-based sensing methods for navigating and interacting with environments in extreme conditions. This research has the potential to be applied in a range of real-world tasks, including search-and-rescue missions in unknown and hazardous environments, autonomous navigation in confined spaces where conventional sensors may fail, and structural or environmental inspection in complex or obstructed areas.

Methods

Hardware

For all experiments reported in the main text, we utilized a prototype Crazyflie 2.1 Brushless drone provided by Bitcraze. The drone features $4 \times 08028-10000KV$ high-quality motors and, with a measured weight of 24.2 g for the drone and 9.2 g for the 350 mAh battery (33.4 g total for the bare platform and battery), achieves a flight endurance of over 10 min. We equipped the drone with two whiskers mounted 5 cm apart on a custom-designed 3D-printed mounting. For the whisker placement-angle study, the mounting included holders that allowed different angular orientations of the whiskers to be tested. For the subsequent experiments, the whiskers were mounted at the selected 45° placement angle. An identical mounting was added on the drone's tail to provide symmetry and stability, with the battery position adjusted to partially compensate for the forward center-of-mass shift introduced by the whisker module, while maintaining a small forward bias that helps reduce clockwise pitch moment during contact interaction. The total additional payload—including the whiskers, two mountings, wiring, and pin headers—added approximately 6 g to the drone. Velocity, altitude control, and odometry were handled using the Flowdeck V2 (PMW3901 optical flow sensor, VL53L1x ToF laser sensor). Additionally, the battery holder deck was replaced with a multi-range deck ($5 \times VL53L1x$ ToF laser sensors) to facilitate comparisons with the whisker's distance estimation performance. In this configuration, the total weight was 44.1 g, and the platform achieved approximately 7 min of flight time when running our pipeline; details are provided in the Supplementary Materials, Flight Time Analysis. Each whisker incorporated three barometers, with temperature compensation and data acquisition managed by an onboard STM32F070F6 microcontroller. This microcontroller communicated with the autopilot via two UART interfaces at 50 Hz.

All other signal preprocessing, state estimation, environmental perception, and control tasks were executed on the autopilot's STM32F4 microcontroller via a customized on-board application at 50 Hz. Logging was performed off-board via Crazyradio PA, and the default Crazyflie firmware served as the autopilot software. State estimation of the drone was performed with the default extended Kalman filter.

Tactile drift online recurrent compensation

We propose the TDORC method to separate contact-related tactile signals from measurement artifacts. The raw signal $S_r(t)$ is modeled as the sum of the desired tactile response and three major artifact components: temperature-induced drift, hysteresis-related post-contact recovery effects, and vibration-induced ringing:

$$S_r(t) = S_t(t) + N_{td}(t) + N_h(t) + N_r(t) \quad (1)$$

where $S_t(t)$ is true signal, $N_{td}(t)$ is noise caused by temperature fluctuations, $N_h(t)$ is noise from partial inelastic strain recovery, $N_r(t)$ is noise induced by drone vibrations or sensor contact events. In free-flight conditions (e.g., hovering), the true signal is assumed to be zero:

$$S_t(t) \approx 0. \quad (2)$$

Thus, during free-flying, the raw signal $S_r(t)$ is dominated by noise components:

$$S_r(t) \approx N_{td}(t) + N_h(t) + N_r(t). \quad (3)$$

To model the temperature drift $N_{td}(t)$, we use a first-order linear regression:

$$N_{td}(t) = a \cdot t + b \quad (4)$$

where t is the time step in every time windows, and a, b are regression parameters, calculated by the least squares method. These parameters are first fitted during the initial free-flying state and used to calibrate the raw signal:

$$S_c(t) = S_r(t) - N_{td}(t) \quad (5)$$

The parameters a (slope) and b (intercept) are computed using the least squares method as follows:

$$a = \frac{n \cdot \sum_{t=0}^n t S_r - (\sum_{t=0}^n t) \cdot (\sum_{t=0}^n S_r)}{n \cdot \sum_{t=0}^n t^2 - (\sum_{t=0}^n t)^2}, \quad (6)$$

$$b = \frac{\sum_{t=0}^n S_r - a \cdot \sum_{t=0}^n t}{n}, \quad (7)$$

where n is the number of data points in the time window, and P_t is the measured pressure value corresponding to time step t .

To maintain accurate noise models over time, we use an adaptive mechanism to recalibrate the noise parameters. If the calibrated signal $S_c(t)$ remains below a predefined threshold, indicating that the true signal is minimal (e.g., no contact forces), the system considers this a free-flight state and recalculates the drift parameters:

$$a, b \leftarrow \text{Re-fit using } S_r(t) \text{ in free-flight state.} \quad (8)$$

To mitigate delayed recovery from hysteresis after contact loss and to suppress high-frequency ringing, we apply a first-order Butterworth

bandpass filter (0.05–1 Hz), implemented as:

$$S_p(t) = b_0 \cdot S_c(t) + z_0, \quad (9)$$

$$z_0 = b_1 \cdot S_c(t) - a_1 \cdot S_p(t) + z_1, \quad (10)$$

$$z_1 = b_2 \cdot S_c(t) - a_2 \cdot S_p(t), \quad (11)$$

where $S_c(t)$ is the temperature-calibrated input, and $S_p(t)$ is the filtered output. The filter coefficients are:

$$b_0 = 0.0564, \quad b_1 = 0, \quad b_2 = -0.0564, \\ a_0 = 1.0, \quad a_1 = -1.8865, \quad a_2 = 0.8872.$$

This structure enables efficient real-time processing of each incoming sample, attenuating low-frequency drift and high-frequency noise to stabilize the tactile signal.

Whisker sensor model

The whisker sensor model maps the pre-processed signals from the three barometers to a 1D contact depth estimate. For each whisker, the input to the model is the normalized three-channel signal vector $\mathbf{s}_n = [S_n^1, S_n^2, S_n^3]^T$, where

$$S_n^i = \frac{S_p^i - \mu^i}{\sigma^i}, \quad i = 1, 2, 3, \quad (12)$$

and μ^i and σ^i are the mean and standard deviation of the training data for the i -th barometer.

We use an MLP to predict depth from $\mathbf{s}_n$. Ground-truth depths are obtained from the relative pose between the drone and the wall measured by the OptiTrack system. Since the platform operates in 2D, the left and right whisker depths are computed geometrically from the relative position and orientation of the drone and the wall. The network is trained by minimizing the mean squared error

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (m_i - d_{t,i})^2, \quad (13)$$

where m_i is the predicted depth and $d_{t,i}$ is the corresponding ground-truth depth for sample i . Optimization is performed using Adam. Network architecture and hyperparameter selection are provided in the Supplementary Materials.

Whiskered drone motion model

To estimate wall distance from tactile sensing, we assume: (1) the contacted obstacle is static in the world frame, (2) each whisker has at most one contact point at a time, and (3) the contacted surface is locally planar.

As shown in Fig. S4, we define the state of the system in the drone frame $\{D\}$ as:

$$\mathbf{x}_k = \begin{bmatrix} d_{\text{wall},k} \\ \theta_{\text{wall},k} \end{bmatrix}, \quad (14)$$

where $d_{\text{wall},k}$ is the perpendicular distance to the wall and $\theta_{\text{wall},k}$ is the wall orientation relative to the drone heading.

The wall orientation is estimated from the left and right whisker depths as:

$$\theta_{\text{wall}}^D = \arctan\left(\frac{d_l - d_r}{s}\right), \quad (15)$$

where d_l and d_r are the estimated depths from the left and right whiskers, and s is the distance between the whiskers. The relative wall

orientation in the drone frame is related to the world-frame quantities by

$$\theta_{\text{wall}}^D = \theta_{\text{wall}}^W - \psi^W. \quad (16)$$

where ψ^W is the drone yaw angle in the world frame. The drone displacement between two consecutive time steps is

$$\Delta p_k = \sqrt{(x_{d,k} - x_{d,k-1})^2 + (y_{d,k} - y_{d,k-1})^2}, \quad (17)$$

and the predicted wall distance is updated as:

$$d_{k+1}^- = \frac{\Delta p_k \sin(\theta_{\text{wall},k}^W - \arctan(\Delta y / \Delta x)) + d_k \cos(\theta_{\text{wall},k}^D)}{\cos(\theta_{\text{wall},k+1}^D)}, \quad (18)$$

where $\Delta x = x_{d,k} - x_{d,k-1}$ and $\Delta y = y_{d,k} - y_{d,k-1}$.

For the simplified process model, we assume $d_l = d_r$, such that $\theta_{\text{wall}}^D = 0$.

Sensor fusion for tactile depth estimation

We fuse the whisker-based depth estimate from the sensor model with the motion-model prediction using a Kalman filter. The prediction step uses Eq. (18), and the covariance is propagated as:

$$\mathbf{P}_{k+1}^- = \mathbf{P}_k + \mathbf{Q}, \quad (19)$$

where $\mathbf{Q}$ is the process noise covariance.

The update step incorporates the sensor-model depth estimate m :

$$\mathbf{K}_k = \mathbf{P}_{k+1}^- (\mathbf{P}_{k+1}^- + \mathbf{R})^{-1}, \quad (20)$$

$$d_{k+1} = d_{k+1}^- + \mathbf{K}_k (m - d_{k+1}^-), \quad (21)$$

$$\mathbf{P}_{k+1} = (\mathbf{I} - \mathbf{K}_k) \mathbf{P}_{k+1}^-, \quad (22)$$

where $\mathbf{R}$ is the measurement noise covariance.

Data collection for tactile depth estimation

To facilitate tactile depth estimation, we developed a structured data collection framework. The drone is positioned at a specified distance from a target wall, oriented perpendicular to the wall's plane. After takeoff, the drone ascends to the designated altitude and hovers momentarily to initialize the TDORC. It then moves forward at a velocity of 20 cm/s under velocity control. Upon detecting contact with an obstacle, a predefined threshold confirms the interaction. If the preprocessed signal surpasses this threshold, the drone moves laterally to the right at 20 cm/s for 2 s, sweeping along the wall to collect training data for distance estimation (Fig. 5a, d). Velocity control is implemented using a PID controller on the Crazyflie platform. Additionally, OptiTrack is employed to capture precise position data for both the drone and the wall, providing GT values for depth estimation. To demonstrate the advantages of our whisker-based approach over a laser sensor—a lightweight, low-cost, and highly accurate emissive device for detecting nearby obstacles—we mounted a time-of-flight (ToF) laser sensor (Crazyflie Multi-ranger deck) on the drone. This sensor measures the wall distance along the direction perpendicular to the drone's heading. To evaluate the proposed tactile depth estimation method, we collected two datasets under different experimental settings:

- Dataset 1: a whiteboard was used as the wall (Fig. 5a), and the state information for velocity control was provided by OptiTrack. A total of 12 flights were conducted, with the first 9 flights used as the training set and the remaining 3 as the test set.

- Dataset 2: a glass wall was used (Fig. 5d), and the drone was equipped with a FlowDeck to provide the state information needed for velocity control. OptiTrack was used solely to obtain GT positions for computing GT tactile depth. A total of 12 flights were performed, with the first 9 flights used as the training set and the remaining 3 as the test set. A video compilation can be found in Supplementary Movie 3.

Whisker-based aerial tactile navigation strategy

We implemented a whisker-based aerial tactile navigation strategy using an onboard finite-state machine (FSM) to control wall-following maneuvers (Fig. 3d, top, pink/red). The FSM operates with the following key parameters: whisker thresholds ($T_{\min}$, $T_{\max}$), maximum velocity ($V_{\max}$), turn rate ($\omega_{\max}$), and state of the inner control loop.

The FSM transitions are as follows:

- Hovering: the drone hovers to initialize tactile drift compensation.
- Flying forward: the drone moves forward at $V_{\max}$.
- Wall following: triggered when either whisker's S_p^{up} exceeds the contact threshold T_c , more details see Fig. S5.
- Resume forward flight: once the obstacle is cleared.
- Landing: executed when the target is reached.

The onboard FSM runs at 50 Hz and directly governs the drone's motion using whisker input. OptiTrack was used to record the drone's absolute position and orientation for post-flight analysis.

Penalty-based active perception in aerial tactile mapping

In this method, a 2D environmental model is reconstructed using GPIS. Corners are identified by applying a potential function with an uncertainty penalty. The point with the highest uncertainty on the contour is then selected for further exploration. The details of the pipeline and algorithmic steps for corner penalty-based active perception as follows (Fig. S5).

Gaussian process implicit surfaces. GPIS⁴⁵ provide a probabilistic framework for reconstructing environmental surfaces and detecting geometric information, particularly from sparse and noisy observations. GPIS defines the surface as the zero-level set of an implicit function, which is modeled as a Gaussian process (GP)⁴⁶. In GPIS, the surface is represented as the zero-level set of a function $f(\mathbf{x})$:

$$S = \{\mathbf{x} \in \mathbb{R}^3 | f(\mathbf{x}) = 0\}, \quad (23)$$

where $f(\mathbf{x})$ is drawn from a GP:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad (24)$$

with mean function $m(\mathbf{x})$ and covariance kernel $k(\mathbf{x}, \mathbf{x}')$. In this work, we employ the Inverse Multiquadric (IMQ) Kernel, which is particularly well-suited for modeling smooth surfaces.

The IMQ kernel is defined as:

$$k_{\text{IMQ}}(\mathbf{x}, \mathbf{x}') = \left(c^2 + \|\mathbf{x} - \mathbf{x}'\|^2 \right)^{-\beta}, \quad (25)$$

where c is a scaling parameter controlling the overall influence of the kernel. $\beta > 0$ determines the decay rate and smoothness of the function. Compared to the commonly used squared exponential (SE) kernel, the IMQ kernel has heavier tails, allowing it to generalize better in regions with sparse data.

Given a set of training points $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N$ and their corresponding signed distance values $\mathcal{Y} = \{y_i\}_{i=1}^N$, the posterior mean $\hat{f}(\mathbf{x}^*)$ and

variance $\hat{\sigma}^2(\mathbf{x}^*)$ at a query point $\mathbf{x}^*$ are computed as:

$$\hat{f}(\mathbf{x}^*) = \mathbf{k}(\mathbf{x}^*, \mathcal{X})^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}, \quad (26)$$

$$\hat{\sigma}^2(\mathbf{x}^*) = k_{\text{IMQ}}(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}(\mathbf{x}^*, \mathcal{X})^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(\mathbf{x}^*, \mathcal{X}), \quad (27)$$

where $\mathbf{K} = k_{\text{IMQ}}(\mathbf{x}_i, \mathbf{x}_j)$ is the covariance matrix of the training points, $\mathbf{k}(\mathbf{x}^*, \mathcal{X})$ is the covariance vector between the query point $\mathbf{x}^*$ and the training points.

Gradient: the gradient of the implicit function, $\nabla f(\mathbf{x})$, describes the rate of change at the query point $\mathbf{x}^*$. To compute the gradient, we take the first derivative of the posterior mean $\hat{f}(\mathbf{x}^*)$:

$$\nabla f(\mathbf{x}^*) = \frac{\partial \hat{f}(\mathbf{x}^*)}{\partial \mathbf{x}^*} = \frac{\partial}{\partial \mathbf{x}^*} \mathbf{k}(\mathbf{x}^*, \mathcal{X})^\top \mathbf{A} \mathbf{y}, \quad (28)$$

where $\mathbf{A} = (\mathbf{K} + \sigma^2 \mathbf{I})^{-1}$ is the inverse covariance matrix.

For the IMQ kernel, the first derivative is given by:

$$\frac{\partial k_{\text{IMQ}}(\mathbf{x}^*, \mathbf{x}_i)}{\partial \mathbf{x}^*} = - \frac{(\mathbf{x}^* - \mathbf{x}_i)}{(\|\mathbf{x}^* - \mathbf{x}_i\|^2 + c^2)^{3/2}}, \quad (29)$$

where c is a scale parameter.

Hessian: to compute the Hessian matrix $\nabla^2 f(\mathbf{x})$, which describes the second-order changes of the implicit surface at $\mathbf{x}^*$, we need to calculate the second derivative of the posterior mean $\hat{f}(\mathbf{x}^*)$. This is given by:

$$\nabla^2 f(\mathbf{x}^*) = \frac{\partial^2 \hat{f}(\mathbf{x}^*)}{\partial \mathbf{x}^{*2}} = \frac{\partial^2}{\partial \mathbf{x}^{*2}} \mathbf{k}(\mathbf{x}^*, \mathcal{X})^\top \mathbf{A} \mathbf{y}. \quad (30)$$

The second derivative of the kernel function is:

$$\frac{\partial^2 k_{\text{IMQ}}(\mathbf{x}^*, \mathbf{x}_i)}{\partial \mathbf{x}^{*2}} = \frac{1}{(\|\mathbf{x}^* - \mathbf{x}_i\|^2 + c^2)^{5/2}} \left[3(\mathbf{x}^* - \mathbf{x}_i)(\mathbf{x}^* - \mathbf{x}_i)^\top - (\|\mathbf{x}^* - \mathbf{x}_i\|^2 + c^2) \mathbf{I} \right], \quad (31)$$

where $\mathbf{I}$ is the identity matrix. The gradients $\nabla f(\mathbf{x})$ and Hessians $\nabla^2 f(\mathbf{x})$ of the implicit function provide geometric information about the surface.

Curvature: the curvature κ at a point $\mathbf{x}^*$ is computed using both the gradient $\nabla f(\mathbf{x}^*)$ and the Hessian matrix $\nabla^2 f(\mathbf{x}^*)$. The curvature is defined as follows:

$$\kappa = \frac{\nabla f(\mathbf{x}^*)^\top \nabla^2 f(\mathbf{x}^*) \nabla f(\mathbf{x}^*) - |\nabla f(\mathbf{x}^*)|^2 \text{Tr}(\nabla^2 f(\mathbf{x}^*))}{|\nabla f(\mathbf{x}^*)|^3}, \quad (32)$$

this equation captures the relationship between the gradient and Hessian to compute the local curvature⁴⁷.

High curvature clustering for corner detection. To identify potentially risky corners, we cluster contour points with high negative curvature, which typically indicate concave inward regions (e.g., the intersection of two walls). These regions pose higher collision risks compared to convex shapes (e.g., cylinders), which curve outward and are easier to follow or avoid.

Given that contour points are labeled as 0 and interior points as -1, we focus on points where curvature $\kappa < q$, with $q < 0$ as a threshold. For a set of contour points $\mathcal{P} = \{\mathbf{p}_i\}_{i=1}^N$ and corresponding curvatures $\mathcal{K} = \{\kappa_i\}_{i=1}^N$, a cluster C_j is defined as:

$$C_j = \{\mathbf{p}_i | \kappa_i < q, i \in [s_j, e_j]\}, \quad (33)$$

where s_j and e_j are the start and end indices. The centroid of each cluster is computed as:

$$\mathbf{c}_j = \frac{1}{|C_j|} \sum_{\mathbf{p}_i \in C_j} \mathbf{p}_i. \quad (34)$$

If the first and last clusters are both high-curvature, we merge them by averaging their centroids:

$$\mathbf{c}'_m = \frac{\mathbf{c}_1 + \mathbf{c}_m}{2}. \quad (35)$$

Exploration strategy: balancing efficiency and safety. Our goal is to enable efficient and safe tactile exploration for exit-seeking drones. Efficiency demands sampling high-uncertainty regions to build a complete surface model, while safety requires avoiding collisions, especially near concave corners.

We use the predictive variance of the GPIS model, $\hat{\sigma}^2(\mathbf{x}^*)$, as an acquisition function:

$$a(\mathbf{x}) = \hat{\sigma}^2(\mathbf{x}^*), \quad (36)$$

which guides the drone toward regions where the model is uncertain.

To ensure safety, we define a repulsive potential function centered at identified corner points $\{\mathbf{c}_j\}$:

$$P(\mathbf{x}) = \min_k \left(-\exp \left(-\frac{\|\mathbf{x} - \mathbf{c}_k\|^2}{2c^2} \right) \right), \quad (37)$$

where c controls the repulsion radius. This discourages the drone from approaching high-risk areas, reducing crash likelihood due to the forward-facing whisker sensor.

The final exploration policy combines both terms:

$$\tilde{a}(\mathbf{x}) = a(\mathbf{x}) + P(\mathbf{x}), \quad (38)$$

where $a(\mathbf{x})$ promotes coverage and $P(\mathbf{x})$ enforces safety. From an optimization perspective, the potential function $P(\mathbf{x})$ introduces an inequality constraint that restricts the exploration range, effectively shaping the drone's trajectory. While $P(\mathbf{x})$ resembles an exploitation term—since it uses prior knowledge of the corner points $\{\mathbf{c}_j\}$ —its primary purpose is to ensure safe exploration rather than directly improving the model. Thus, $P(\mathbf{x})$ functions as a safety-driven exploitation mechanism, balancing the dual objectives of efficient exploration and collision avoidance.

Active aerial tactile exploration

As shown in Fig. 3d, the drone alternates between wall-following and GPIS-based uncertainty reduction. It initially explores in four orthogonal directions (90° apart) to build a coarse environmental model. Upon contact with a surface, it follows the wall for 2 seconds to gather contour data, then flies backward for 1 second to collect interior points. To enable onboard processing, we reduce memory usage by downsampling—storing four data points per wall-follow and one per retreat. We then apply a corner-penalty exploration module that combines GPIS uncertainty with high-curvature detection to discourage entry into concave, high-risk areas. These corners are implicitly identified by analyzing the curvature of the reconstructed contours. The exit is not predefined; instead, it naturally emerges once all contour segments have been explored. The corner-aware exploration strategy accelerates mapping by steering the drone toward uncertain areas while avoiding dangerous zones. This ensures safe, efficient, and complete tactile exploration in unknown, confined environments.

Simulations were conducted using OmniDrone⁴⁸ within Isaac Sim in a confined environment with a narrow exit. The Hummingbird drone

dynamic model was used with Lee-controller-based⁴⁹ velocity and attitude rate control for movement and steering. Two forward-facing single-pixel LiDAR sensors approximate whisker functionality, with Gaussian noise (mean 0, variance 1 cm) simulating real-world whisker noise. The simulation validates the feasibility of the active tactile exploration algorithm and corner-penalty strategy.

To facilitate quantitative evaluation, we implement several baseline and ablation variants of the proposed exploration strategy in simulation. Specifically, we consider: (1) *w/o Corner Penalty*, in which the curvature-based penalty is removed; (2) *w/o Sweeping*, in which the wall-following sweeping behavior is disabled; in this case, the corner-penalty term is also inactive, since corner avoidance is only needed during sweeping-based target selection; and (3) *Random Exploration*, in which the drone selects exploration directions without leveraging GPIS uncertainty or geometric cues. All methods share the same initialization procedure, where the drone first explores in four orthogonal directions to build a coarse environmental model.

We evaluate performance using three groups of metrics.

Exit success rate is defined as the percentage of trials in which the drone successfully locates the exit and flies out of the environment.

Reconstruction accuracy is quantified using Chamfer Distance (CD) and Root Mean Square Distance (RMSD) between the reconstructed contour and the ground-truth contour. Let $\mathcal{P} = \{p_i\}$ and $\mathcal{G} = \{g_i\}$ denote the sampled points on the reconstructed and ground-truth contours, respectively. The Chamfer Distance is defined as:

$$CD(\mathcal{P}, \mathcal{G}) = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \min_{g \in \mathcal{G}} \|p - g\|_2 + \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \min_{p \in \mathcal{P}} \|g - p\|_2, \quad (39)$$

where $|\mathcal{P}|$ and $|\mathcal{G}|$ are the numbers of sampled points in the reconstructed and ground-truth contours, respectively, and $\|\cdot\|_2$ denotes the Euclidean norm. CD measures the bidirectional average nearest-neighbor distance between the two contours.

The RMSD is defined as:

$$RMSD(\mathcal{P}, \mathcal{G}) = \sqrt{\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} d(p, \mathcal{G})^2}, \quad (40)$$

where $d(p, \mathcal{G}) = \min_{g \in \mathcal{G}} \|p - g\|_2$. In our implementation, this distance is computed as the minimum point-to-segment distance from p to the piecewise-linear ground-truth contour.

Exploration efficiency is measured by the total travel distance and the sweeping count. The travel distance is computed as the accumulated Euclidean distance along the drone trajectory. The sweeping count denotes the number of sweeping interactions performed during exploration. To ensure fair comparison, exploration-efficiency metrics are computed only over trials that are not prematurely terminated by bumping events, whereas reconstruction metrics are reported over all trials.

Data availability

The simulation and real-world experimental data generated in this study are provided as Supplementary Data 1 and are also available in the GitHub repository at <https://github.com/BioMorphic-Intelligence-Lab/Whiskered-drone>, which has been archived on Zenodo with the <https://doi.org/10.5281/zenodo.21641894>⁵⁰. Source data are provided with this paper.

Code availability

The code used in this study is publicly available in the GitHub repository: <https://github.com/BioMorphic-Intelligence-Lab/Whiskered-drone>, which has been archived on Zenodo with the <https://doi.org/10.5281/zenodo.21641894>⁵⁰. The repository includes the simulation code, the modified Crazyflie firmware for the real-world platform, the

onboard tactile navigation and exploration implementation, and the whisker depth estimation code.

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Conceptualization: S.Hamaza and C.Ye. Methodology: C.Ye, G.C.H.E.d.C., and S.Hamaza. Software and Hardware: C.Ye. Experiments: C.Ye. Visuals: C.Ye and S.Hamaza. Supervision: G.C.H.E.d.C. and S.Hamaza. Funding: S.Hamaza. Writing of the original draft: C.Ye. Draft revision and editing: C.Ye, G.C.H.E.d.C., and S.Hamaza.

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Competing interests

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Additional information

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